

Testing C# Code



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Overview



Testing your application using the debugger

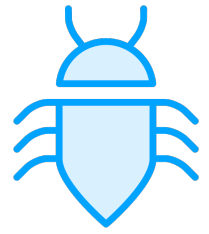
Writing a unit test



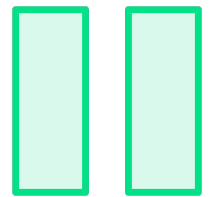
Testing Your Application Using the Debugger



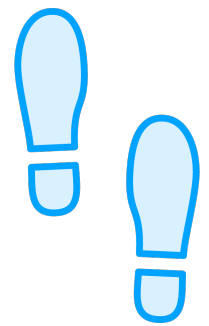
Using the Debugger



Run in Debug mode



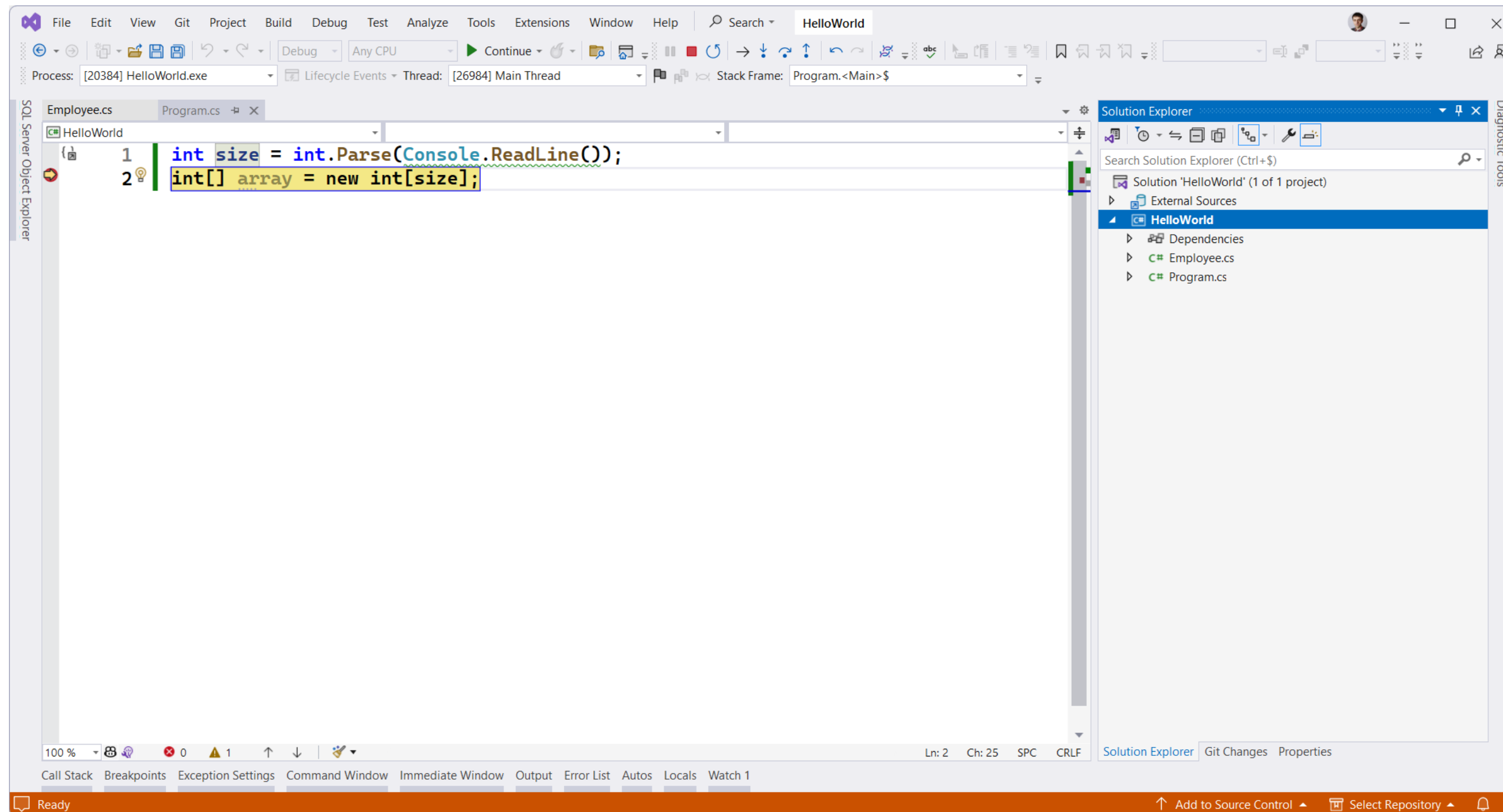
Breakpoints



Stepping through the code



Using Break Mode and Breakpoints



Debugger Commands

F5: start debugging

F11: Step Into

F10: Step over

Shift + F11: Step out



Demo



Using the debugger

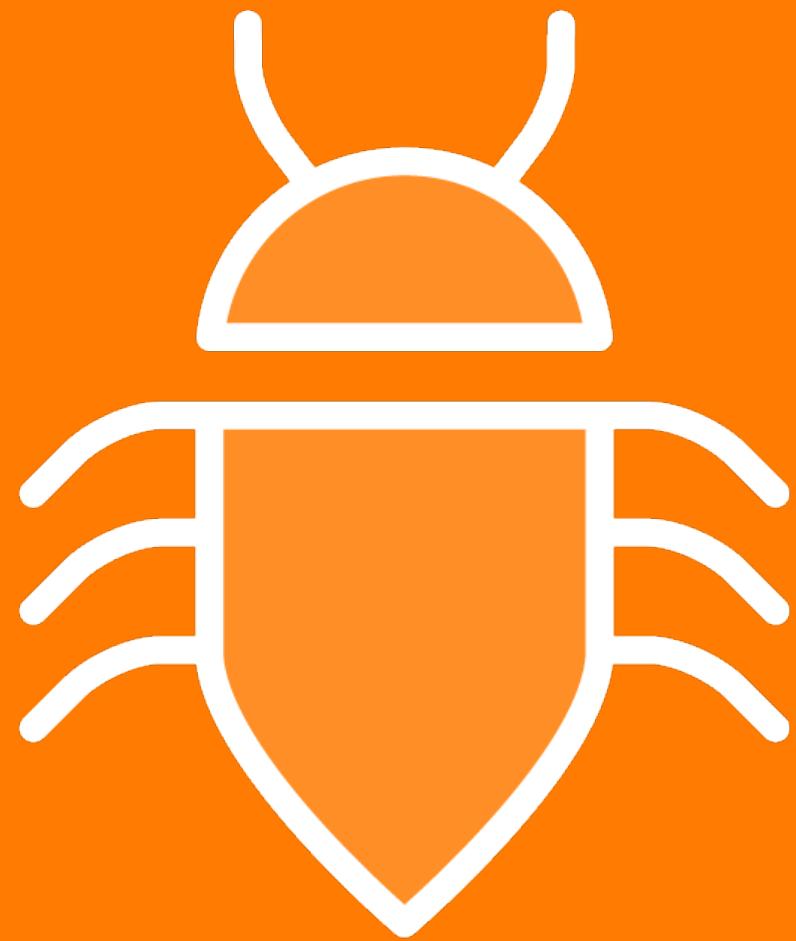
Understanding the debugger windows





Writing a Unit Test





We aren't perfect...

Introducing new bugs is easy!

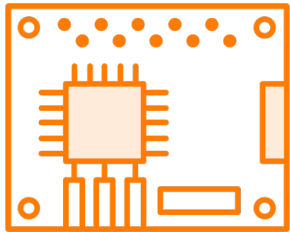
**We can try harnessing our code
using a unit test**



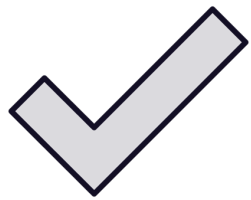
Introducing Unit Tests



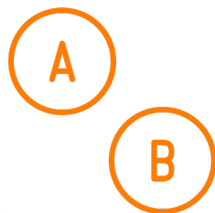
Code to test other code



Small components of the application



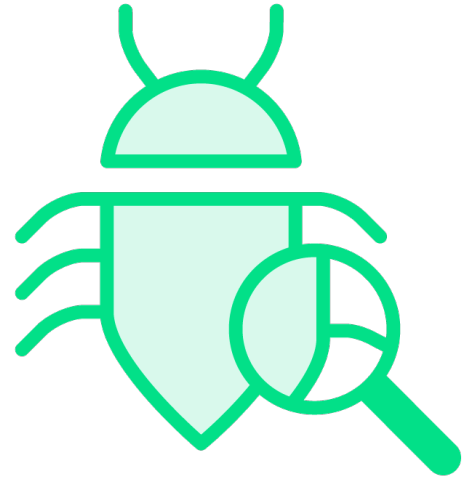
Validate value



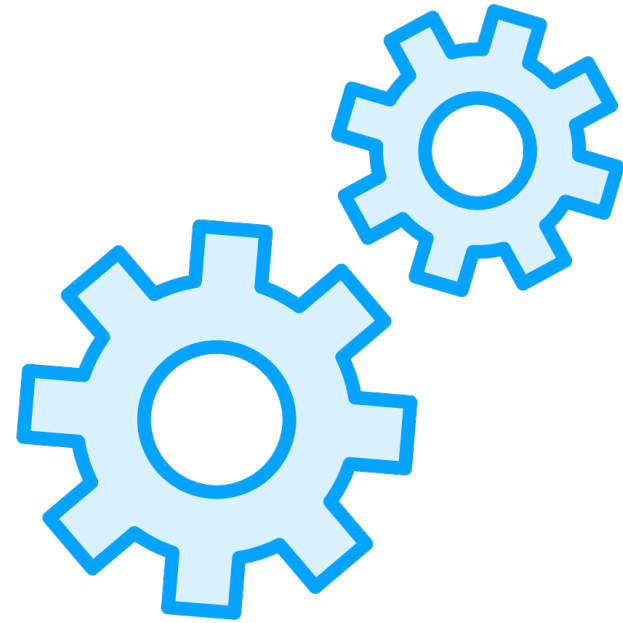
Isolate part of the code



Advantages of Unit Tests



Find bugs



**Change without
fear of breaking
something**



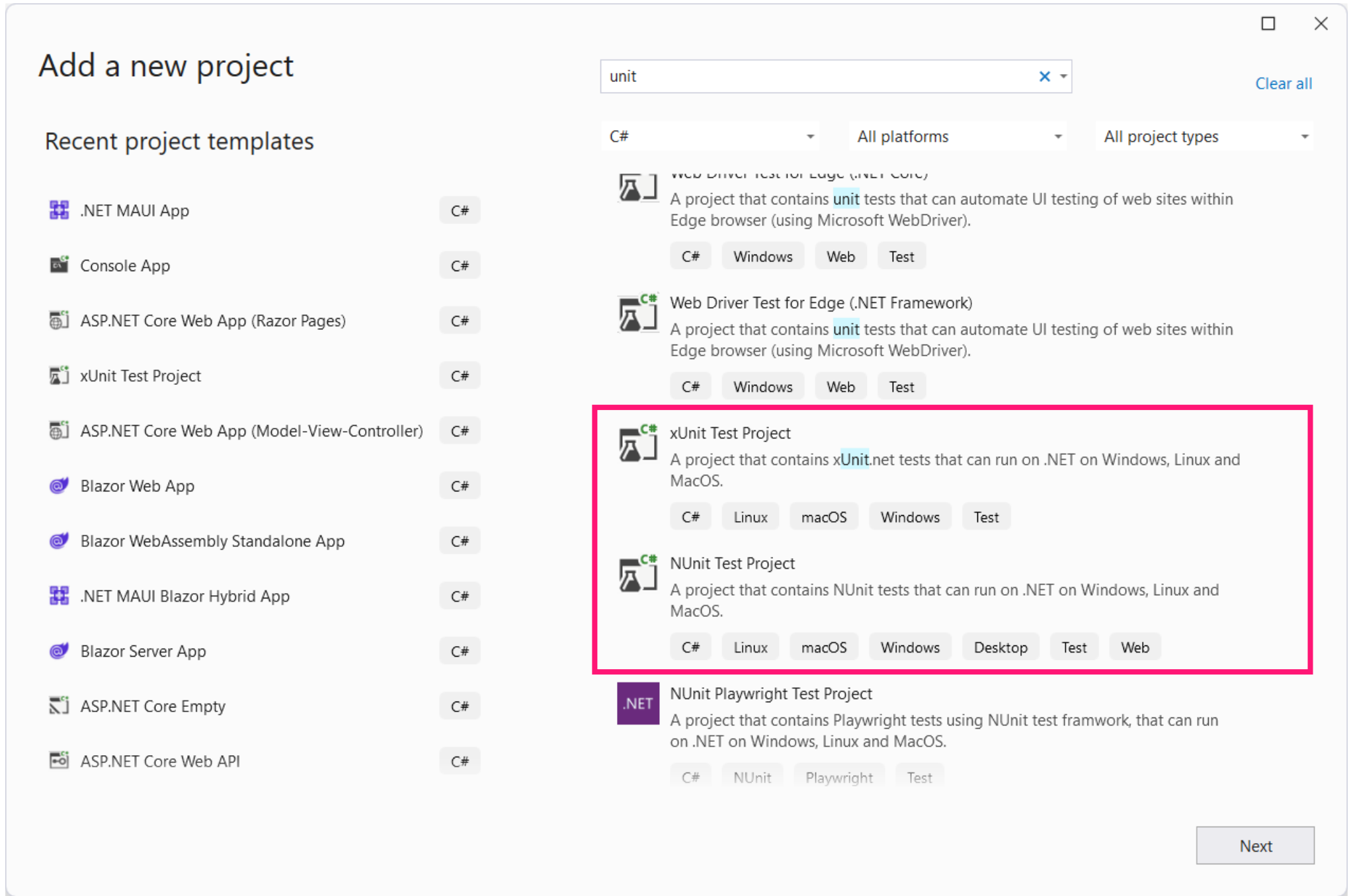
Improve quality



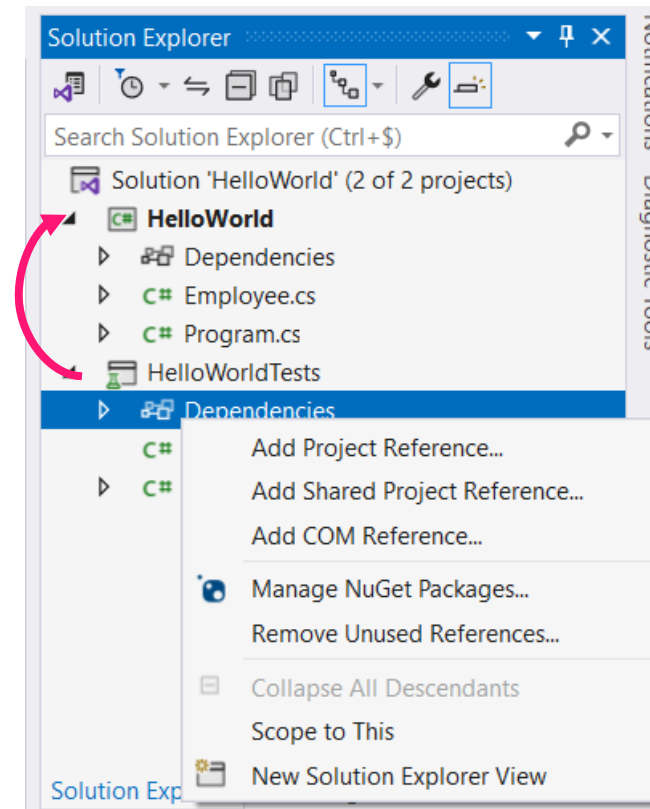
**Documentation
of the code**



Creating a Unit Test Project



Sidestep: Adding a Reference



Structure of a Unit Test

Arrange

Act

Assert



Writing a Unit Test

```
public class EmployeeTests
{
    [Fact]
    public void PerformWork_Adds_DefaultNumberOfHours_IfNoValueSpecified()
    {
        //Arrange
        Employee employee = new Employee(...);

        //Act
        employee.PerformWork();

        //Assert
        Assert.Equal(1, employee.NumberOfHoursWorked);
    }
}
```



Running Tests with Test Explorer

Test Explorer

▶▶

↺

✖

🔍

🧪 2

✅ 2

❌ 0

📄

⚡

⌵

📄

⚙️

Ready

Test	Duration	Traits
▶️✅ BethanysPieShopHRM.Tests (2)	4 ms	
▶️✅ BethanysPieShopHRM.Tests (2)	4 ms	
▶️✅ EmployeeTests (2)	4 ms	

Group Summary

BethanysPieShopHRM.Tests

Tests in group: 2

🕒 Total Duration: 4 ms

Outcomes

✅ 2 Passed



Demo



Creating a unit test project

Adding a unit test

Running the test using Test Explorer



Summary



Using the debugger, we can test our code and inspect values

Unit tests can help with making code more resistant to errors being introduced



Up Next:

Working with files

